

DES Block Cipher

Module II

DES Block Cipher

The Data Encryption Standard (DES):

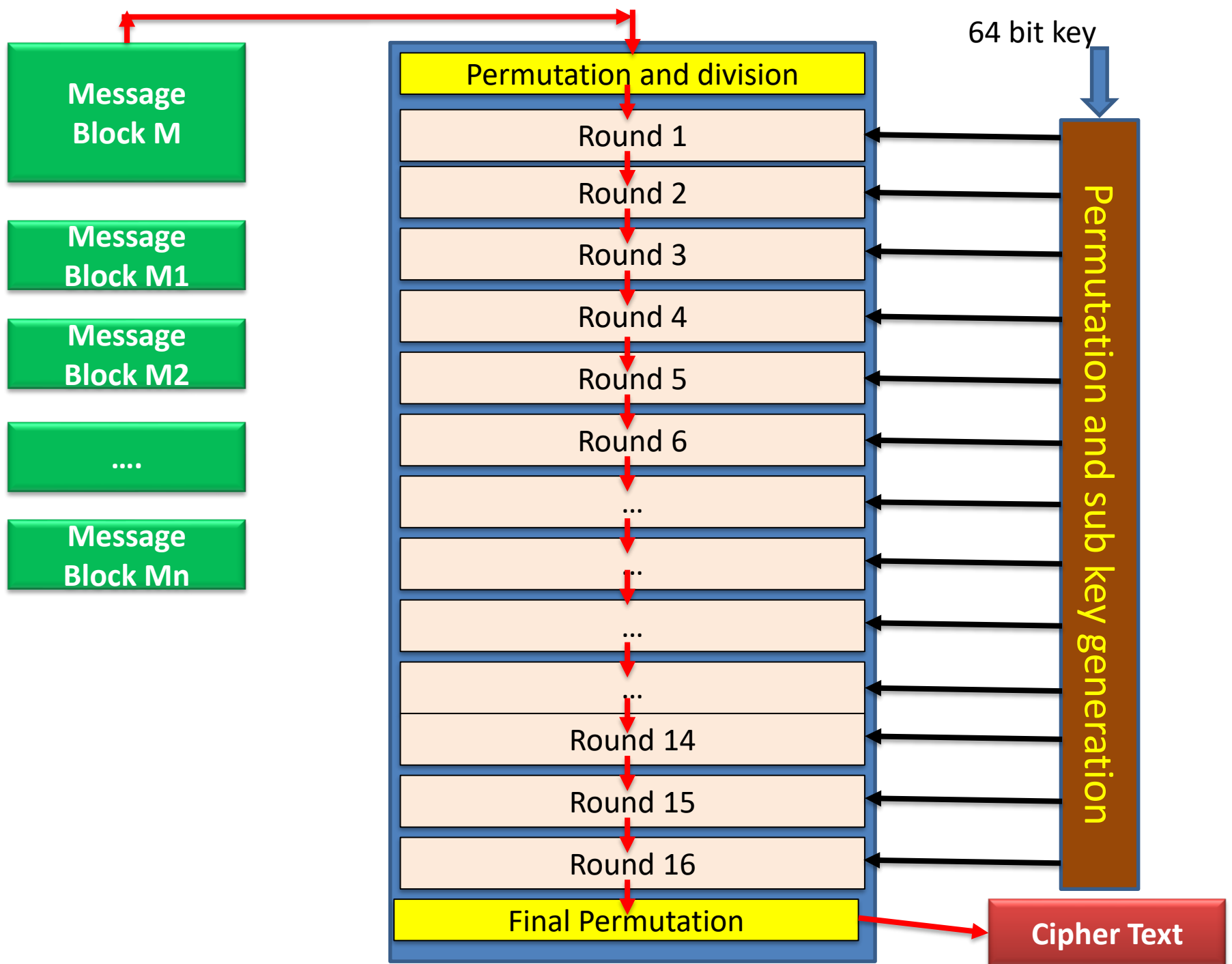
This algorithm adopted in 1977 by the National Institute of Standards and Technology (**NIST**). The algorithm itself is referred to as the Data Encryption Algorithm (**DEA**). For DES, data are encrypted in **64-bit blocks** using a **56-bit key**. The algorithm transforms 64-bit input in a series of steps into a **64-bit output**. The same steps, with the same key, are used to reverse the encryption.

DES Block Cipher

DES encryption algorithm:

The general structure of the DES consists of

- (1) key schedule,**
- (2) round function**
- (3) Initial and final permutation.**



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Step1: Plaintext is broken into blocks of length **64 bits**.

Step2: The **64-bit** block undergoes an initial permutation (IP) using initial permutation **IP** table, $IP(M)$.

Step3: The **64-bit** permuted input is divided into **two 32-bit** blocks: left (**L**) and right (**R**). The initial values of the left and right blocks are denoted L_0 and R_0 .

Step4: There are **16** rounds of operations on the **L** and **R** blocks. During each round, the following formula is applied:

$$L_n = R_{n-1}$$

$$R_n = L_{n-1} \text{ XOR } F(R_{n-1}, K_n)$$

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Step5: The function **F(.)** represents the heart of the DES algorithm. This function implements the following operations:

1. Expansion
2. Key mixing
3. Substitution
4. Permutation

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Step5:

1-Expansion: The right **32-bit** half-block is expanded to **48 bits** using the **expansion permutation (E)** table, $E(R_{n-1})$.

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Step5:

2-Key mixing: The expanded result is combined with a **subkey** using an XOR operation. Sixteen 48-bit subkeys (one for each round) are derived from the main key using the **key schedule**, $K_n + E(R_{n-1})$.

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Step5:

3-Substitution: After mixing in the subkeys, the block is divided into eight 6-bit pieces and fed into the substitution boxes (S-boxes), which implements nonlinear transformation. Each 6-bit piece uses as an address in the **S-boxes** where the first and last bits are used to address the i^{th} row and the middle four bits to address the j^{th} column in the S-boxes. The output of each **S-box** is 4-bit length piece. The output of all **eight S-boxes** is then combined into **32** bit section.

$$K_n + E(R_{n-1}) = B_1 B_2 B_3 B_4 B_5 B_6 B_7 B_8$$

$$S(K_n + E(R_{n-1})) = S1(B_1) S2(B_2) S3(B_3) S4(B_4) S5(B_5) S6(B_6) S7(B_7) S8(B_8)$$

DES Block Cipher

Step5:

4-Permutation: The **32** bits outputs from the S-boxes are rearranged using the **P-box**, $F = P(S(K_n + E(R_{n-1})))$

DES Block Cipher

Step6: The results from the final DES round (i.e., L_{16} and R_{16}) are recombined into a 64-bit value and rearranged using an inverse initial permutation (IP^{-1}) table. The output from IP^{-1} is the 64-bit ciphertext block.

Generating 16 Sub keys

The 64 bit key is permuted according to the following PC-1 Table. Only 56 bits of the original key appear in the permuted key.

K = 00010011 00110100 01010111 01111001 10011011 10111100 11011111 11110001



57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	21	4

We get the 56 bit key permutation

PC1 Table

K₊ = 1111000 0110011 0010101 0101111 0101010 1011001 1001111 0001111

Generating 16 Sub keys

We get the 56 bit key permutation

$K^+ = 1111000\ 0110011\ 0010101\ 0101111\ 0101010\ 1011001\ 1001111\ 0001111$

Split this key into left and right halves, CO and DO, where each half has 28 bits.

From the permuted key K^+ , we get

CO	1111000	0110011	0010101	0101111
DO	0101010	1011001	1001111	0001111

Generating 16 Sub keys

From the previous step,

CO 1111000 0110011 0010101 0101111
DO 0101010 1011001 1001111 0001111

Schedule of left shifts.
(Circular left shift)

Iter	No. of left shifts	Iter	No. of left shifts
1	1	11	2
2	1	12	2
3	2	13	2
4	2	14	2
5	2	15	2
6	2	16	1
7	2		
8	2		
9	1		
10	2		

Generating 16 Sub keys

From the previous step,

CO 1111000 0110011 0010101 0101111
DO 0101010 1011001 1001111 0001111

Iteration 1

$C_1 = 1110000110011001010101011111$
 $D_1 = 1010101011001100111100011110$

Iteration 2

$C_2 = 1100001100110010101010111111$
 $D_2 = 0101010110011001111000111101$

Generating 16 Sub keys

From the previous step,

$C_2 = 1100001100110010101010111111$

$D_2 = 0101010110011001111000111101$

Iteration 3



$C_3 = 0000110011001010101011111111$

$D_3 = 0101011001100111100011110101$

Iteration 4



$C_4 = 0011001100101010101111111100$

$D_4 = 0101100110011110001111010101$

Generating 16 Sub keys

From the previous step.

$C_4 = 0011001100101010101111111100$
 $D_4 = 0101100110011110001111010101$

Iteration 5

$C_5 = 1100110010101010111111110000$
 $D_5 = 0110011001111000111101010101$

Iteration 6

Generating 16 Sub keys

$C_6 = 0011001010101011111111000011$
 $D_6 = 1001100111100011110101010101$

$C_7 = 1100101010101111111100001100$
 $D_7 = 0110011110001111010101010110$

$C_8 = 0010101010111111110000110011$
 $D_8 = 1001111000111101010101011001$

$C_9 = 0101010101111111100001100110$
 $D_9 = 0011110001111010101010110011$

$C_{10} = 01010101111111110000110011001$
 $D_{10} = 1111000111101010101011001100$

$C_{11} = 01010111111111000011001100101$
 $D_{11} = 1100011110101010101100110011$

$C_{12} = 0101111111100001100110010101$
 $D_{12} = 0001111010101010110011001111$

$C_{13} = 0111111110000110011001010101$
 $D_{13} = 0111101010101011001100111100$

$C_{14} = 1111111000011001100101010101$
 $D_{14} = 1110101010101100110011110001$

$C_{15} = 1111100001100110010101010111$
 $D_{15} = 1010101010110011001111000111$

$C_{16} = 1111000011001100101010101111$
 $D_{16} = 0101010101100110011110001111$

Generating 16 Sub keys

We formed the keys K_n , for $1 \leq n \leq 16$, by applying the following permutation table to each of the concatenated pairs $C_n D_n$.

Each pair has 56 bits, but **PC-2** only uses 48 of these.

$C_1 = 1110000110011001010101011111$

$D_1 = 1010101011001100111100011110$

$C_1 D_1$

11100001100110010101010111111010101011001100111100011110

The same process will be repeated to generate $K_2, K_3 \dots K_{16}$ from $C_2 D_2, C_3 D_3 \dots C_{16} D_{16}$.

14	17	11	24	PC-2 1	5
3	28	15	6	21	10
23	19	12	4	26	8
16	7	27	20	13	2
41	52	31	37	47	55
30	40	51	45	33	48
44	49	39	56	34	53
46	42	50	36	29	32

$K_1 = 000110 \ 110000 \ 001011 \ 101111 \ 111111 \ 000111 \ 000001 \ 110010$

Generating 16 Sub keys

$K_1 = 000110\ 110000\ 001011\ 101111\ 111111\ 000111\ 000001\ 110010$
 $K_2 = 011110\ 011010\ 111011\ 011001\ 110110\ 111100\ 100111\ 100101$
 $K_3 = 010101\ 011111\ 110010\ 001010\ 010000\ 101100\ 111110\ 011001$
 $K_4 = 011100\ 101010\ 110111\ 010110\ 110110\ 110011\ 010100\ 011101$
 $K_5 = 011111\ 001110\ 110000\ 000111\ 111010\ 110101\ 001110\ 101000$
 $K_6 = 011000\ 111010\ 010100\ 111110\ 010100\ 000111\ 101100\ 101111$
 $K_7 = 111011\ 001000\ 010010\ 110111\ 111101\ 100001\ 100010\ 111100$
 $K_8 = 111101\ 111000\ 101000\ 111010\ 110000\ 010011\ 101111\ 111011$
 $K_9 = 111000\ 001101\ 101111\ 101011\ 111011\ 011110\ 011110\ 000001$
 $K_{10} = 101100\ 011111\ 001101\ 000111\ 101110\ 100100\ 011001\ 001111$
 $K_{11} = 001000\ 010101\ 111111\ 010011\ 110111\ 101101\ 001110\ 000110$
 $K_{12} = 011101\ 010111\ 000111\ 110101\ 100101\ 000110\ 011111\ 101001$
 $K_{13} = 100101\ 111100\ 010111\ 010001\ 111110\ 101011\ 101001\ 000001$
 $K_{14} = 010111\ 110100\ 001110\ 110111\ 111100\ 101110\ 011100\ 111010$
 $K_{15} = 101111\ 111001\ 000110\ 001101\ 001111\ 010011\ 111100\ 001010$
 $K_{16} = 110010\ 110011\ 110110\ 001011\ 000011\ 100001\ 011111\ 110101$

Generating 16 Sub keys

In summary, In the key schedule, PC1 table reducing the size of 64 bit key into 56 bit key and PC2 table further reducing the size of the 56 bit key into 48 bit key, whereas the Left shift table is used to generate 16 sub keys from a 56 bit key.

